

Configuring pfSense in Dual WAN Failover Mode

This is the second article in series on pfSense. While the introductory article covered pfSense basics, here, readers are taken a step further in their journey of mastering the tool.



The first article in this series included the basic pfSense set-up, installing/configuring the Squid Proxy server and the SquidGuard proxy filter. This article takes off from the stage after pfSense has been configured. It continues further to set up dual WAN failover. Please refer to the earlier article that was carried in the June 2016 issue of *OSFY* to learn more on how to establish a basic firewall.

The new pfSense minor release

Between writing the first and second articles, two pfSense releases have been made available.

- a. 2.3 update 1 provides security patches to mitigate risks from NTP CVEs for FreeBSD. Details of this CVE are available at <https://www.freebsd.org/security/advisories/FreeBSD-SA-16:16.ntp.asc>. Apply this update before continuing further. Being a minor release, there will be no change in the pfSense version number. Once the update is complete, do not forget to restart the NTP service from the *Status – Services* menu. A reboot is not required.
- b. Important changes in version 2.3.1 include bug fixes from the earlier release, fixing of issues released in FreeBSD security advisories related to OpenSSL, atkbd and sendmsg, and the upgradation of OpenVPN version from 2.3.10 to 2.3.11. To apply this update, click on the icon under *Status–Dashboard–System Information – Version* and select *Confirm*. An auto installation process will complete the update and reboot pfSense. Time required for this update will primarily depend upon the speed of your internet connection.

What is a CVE?

As mentioned on cve.mitre.org, common vulnerabilities and exposure (CVE) is a list of information security vulnerabilities and exposures that aims to provide common names for publicly known cyber security issues. The goal of CVE is to make it easier to share data across separate vulnerability capabilities (tools, repositories and services) with this 'common enumeration'.

In laymen's terms, CVEs provide information about vulnerabilities in operating systems, application software and services. These vulnerabilities need to be fixed by their respective vendors to mitigate the risks to users and to improve security.

A background to pfSense dual WAN failover

Many organisations have a primary high speed Internet connection, and a low speed standby connection. Typically, the latter is used if the primary connection fails. The Internet connectivity via the primary connection resumes once it is online again. Configuring pfSense in multi-WAN failover mode automates this task.

Many articles on the Web and pfSense forums have described the installation of pfSense dual WAN failover. Different versions of pfSense may require different set-up instructions. If the steps are not followed meticulously, failover does not work, thus frustrating the administrator/installer.

For the latest pfSense 2.3 release, dual WAN failover configuration is the simple five-step process described below. A few best practices are listed after that.